CreatorPilot

Creator Growth Analysis Application: an AI Growth Coach for Content Creators

1. Overview

Most existing tools (Social Blade, VidIQ, TubeBuddy) only show dashboards and analytics. CreatorPilot is different because it works as a **proactive AI coach**: a Siri-like assistant that watches the creator over time, talks to them, and nudges them to act.

2. Core Concept

The creator connects their accounts (YouTube, Instagram, etc.) through official APIs. The assistant then:

- **Monitors** their upload activity, consistency and performance.
- **Analyzes** what is working and what is not.
- **Guides** them with trending topics and timing advice.
- **Notifies** them at the right moment, like a personal manager.

3. Features

3.1 Conversational AI Coach (chatbot and voice)

- Ask things like "Why did my last video flop?" or "What should I post this week?"
- Voice mode (speech-to-text and text-to-speech), so it feels like Siri for creators.
- The assistant remembers the creator's niche, goals and past advice.

3.2 Smart Activity Monitoring and Notifications

- **Upload gap alert:** "You haven't posted in 6 days. Your audience engagement drops after 7."
- **Best-time alert:** "Your audience is most active at 7 PM today. Post now."
- **Streak tracker:** Rewards for consistency, with gentle reminders when a streak is at risk.
- **Burnout detector:** Notices a falling posting frequency and suggests lighter content or a break.

3.3 Trend Intelligence Engine

- Pulls trending topics, hashtags and sounds in the creator's niche.
- **Trend-fit score:** "This trend matches your niche at 82%. Make a video on it within 48 hours."
- **Trend expiry timer:** Shows how long a trend is likely to stay useful.

3.4 Content Performance Analysis (the ML part)

- **Title and thumbnail scorer:** Predicts the click potential of a title before posting.
- **Sentiment analysis of comments:** Shows what the audience loves, dislikes and asks for.
- **Video drop-off analysis:** Suggests where viewers leave and how to fix the hook.
- **Growth prediction:** Forecasts followers or subscribers for the next 30 days using a time-series model.

3.5 Unique Features That Set the Project Apart

- **Content Gap Finder:** Finds topics the audience asks about in comments that the creator has not covered yet.
- **Weekly Growth Plan generator:** An auto-made plan covering what to post, when, and with what hashtags.
- **Competitor comparison:** Compares the creator with similar-sized creators and shows what they do differently.

- **Idea Vault:** Saves and ranks content ideas by predicted performance.
- **Growth Health Score:** One score (0-100) combining consistency, engagement and trend alignment.
- **Collab Matcher:** Suggests creators in a similar niche and size to collaborate with.
- **Monetization readiness tracker:** Shows progress towards monetization requirements (for example, subscriber and watch-time targets).

4. Suggested Tech Stack

Layer	Options
Frontend	React / Flutter (mobile app)
Backend	Node.js or Python (FastAPI / Flask)
Database	MongoDB / PostgreSQL / Firebase
AI / NLP	LLM API for the chatbot, NLTK / Hugging Face for sentiment
ML	Scikit-learn / Prophet for growth forecasting
Data sources	YouTube Data API, Google Trends (pytrends), Instagram Graph API
Notifications	Firebase Cloud Messaging, scheduled jobs (Celery / cron)
Voice	Web Speech API / Google Speech-to-Text

5. Suggested Modules for the Project Report

1. User and account management
2. Data collection (API integration)
3. Analytics and visualization dashboard
4. Trend analysis module
5. ML prediction module
6. Chatbot and voice assistant
7. Notification and scheduling engine

6. Tips for a Strong Final-Year Project

- **Start with one platform.** YouTube has the easiest free API; expand later.
- **Pick 2 or 3 hero features** (for example, the AI coach, trend-fit score and growth prediction) and build them well rather than building everything.
- Show a **live demo with real channel data**, which impresses evaluators.
- Mention future scope: more platforms, multilingual support, a creator community.